

RESEARCH ARTICLE

Enhancing Intelligent Control Strategies for UAVs: A Comparative Analysis of Fuzzy Logic, Fuzzy PID, and GA-Optimized Fuzzy PID Controllers

NIGATU WANORE MADEBO^{ID}

Information Network Security Administration (INSA), Addis Ababa 124498, Ethiopia

e-mail: nigatu.wanore2015@gmail.com

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ABSTRACT This paper presents advanced control strategies to enhance the stability and trajectory tracking performance of quadrotor systems. The study investigates three control methodologies: the Fuzzy Logic Controller (Fuzzy), the Fuzzy Proportional-Integral-Derivative (FPID) controller, and the Genetic Algorithm (GA)-optimized Fuzzy PID controller (GAFPID). The Fuzzy controller leverages heuristic rules for adaptive control, while the FPID controller integrates conventional PID dynamics with fuzzy logic to improve precision and robustness. The GAFPID controller employs evolutionary computation through a genetic algorithm to optimize parameter tuning, offering superior control performance. Comparative simulations are conducted under diverse operating conditions, including external disturbances and parameter variation scenarios, with performance evaluated using the Integral of Time-weighted Absolute Error (ITAE) metric. Results demonstrate that the GAFPID controller outperforms the other approaches in terms of precision, adaptability, and robustness, establishing it as a promising solution for complex quadrotor applications.

INDEX TERMS FPID, GAFPID, fuzzy, unmanned aerial vehicle (UAV).

I. INTRODUCTION

Quadrotors is a class of unmanned aerial vehicles (UAVs), that has garnered considerable attention owing to their versatility and diverse applications, such as surveillance, environmental monitoring, and delivery services [1], [2]. However, their inherently unstable dynamics and high sensitivity to external disturbances present significant challenges in control system design. Consequently, the development of robust and efficient control strategies is essential to guarantee the stability and optimal performance of quadrotor systems.

Classical Proportional-Integral-Derivative (PID) controllers are extensively utilized in control systems due to their simplicity and effectiveness. Nevertheless, their performance can deteriorate in nonlinear or highly dynamic environments, such as those experienced by quadrotors. To overcome these limitations, fuzzy logic-based controllers have been introduced. Fuzzy Logic Controllers (FLCs) are particularly effective in managing uncertainties and nonlinearities, making them well-suited for complex systems

like quadrotors [3]. The integration of fuzzy logic with traditional PID controllers results in Fuzzy PID controllers, which leverage the benefits of heuristic decision-making and classical control to achieve improved performance [4], [5], [6].

Despite significant advancements, achieving optimal parameter tuning for Fuzzy PID controllers remains a challenge, especially under varying operational conditions. Genetic Algorithms (GAs), a form of evolutionary computation, have demonstrated effectiveness in addressing this challenge by efficiently optimizing control parameters [7], [8], [9]. GA-optimized PID controllers offering enhanced adaptability and robustness in dynamic environments [10], [11], [12], [13], [14], [15], [16]. The distributed formation control of under-actuated unmanned vehicles, accounting for uncertainties and disturbances, has been effectively studied by the authors in [17], [18], and [19].

A. RELATED WORK

Control of quadrotor UAVs has progressed as researchers tackle challenges arising from its nonlinear and inherently

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unstable dynamics. Traditional methods, such as Proportional-Integral-Derivative (PID) controllers, are widely used due to their simplicity; however, they lack the robustness and adaptability required for complex, real-world environments where system parameters and external disturbances vary unpredictably. Reference [20] introduced Takagi-Sugeno fuzzy logic technique for quadrotors the designed controllers are efficient, reliable, robust and flexible than the classical controllers. Author in Reference [21] introduces a robust intelligent control strategy tailored for a highly nonlinear and underactuated quadrotor UAV, considering external disturbances, system uncertainties, and stochastic perturbations. The control framework employs the Mamdani fuzzy logic control technique, addressing the six degrees of freedom (6DOF), the result is validated through simulations, demonstrating efficiency, reliability, robustness, and flexibility. The author in [22] presents a robust control strategy for quadrotor drones, focusing on their nonlinear and multi-input multi-output dynamics. The method combines a fast terminal sliding mode controller with Takagi-Sugeno fuzzy logic to improve control performance. A triangular observer estimates unobservable states, while fuzzy logic reduces sliding mode chattering and mitigates external disturbances. Particle Swarm Optimization further enhances control accuracy by optimizing fuzzy logic coefficients, ensuring stability using the Lyapunov method. Simulations demonstrate the robustness and effectiveness of the proposed strategy under varying conditions. In reference [23], an adaptive fuzzy tracking control scheme is proposed for a quadrotor unmanned aerial vehicle, integrating Lyapunov stability theory with the backstepping technique. Initially, a fuzzy logic system approximates the unmodeled dynamics of the quadrotor. Subsequently, command filtering is employed to calculate the derivatives of the virtual control signals, circumventing the need for complex analytic derivation. Additionally, a nonlinear disturbance observer is introduced to counteract external disturbances and the approximation error inherent in the fuzzy logic system. The implementation of Lyapunov stability analysis demonstrates that the proposed controller effectively stabilizes the quadrotor system, achieving high control accuracy. Experimental results are presented to validate the proposed control strategy. In reference [24], data analysis of quadcopter dynamic attitude on a circular trajectory is presented, specifically comparing the modeling results of conventional Proportional Integral Derivative (PID) and Fuzzy-PID controllers. Simulations examining attitude stability for both control systems were conducted using the Simulink toolbox in MATLAB, allowing clear identification of each control system's performance. The algorithms for each control system, specifically related to roll and pitch angles that influence horizontal movement along a circular trajectory, are explained in detail. The results highlight the impact of tuning variables on the output movement, facilitating a comparison of error magnitudes against reference angles. To enhance the analysis, wind disturbances along each axis were introduced to the

model, making the distinctions between the two control systems more pronounced. Simulation results indicate that the Fuzzy-PID controller exhibits relatively smaller errors compared to the PID controller and demonstrates superior disturbance rejection capabilities. In reference [25], the authors explore trajectory tracking, which is a prevalent application for quadcopters and poses significant research challenges due to the complex dynamics associated with these vehicles. Various control algorithms have been established to stabilize quadcopters along specific trajectories, as following a designated path can be particularly difficult. This paper presents a fuzzy-PID controller designed to stabilize a quadcopter over a specified trajectory, utilizing speed information as an input. The performance of the proposed controller is compared against that of a PD controller and a fuzzy logic controller (FLC) using MATLAB/Simulink to evaluate effectiveness. The simulation results indicate that the fuzzy-PID controller demonstrates superior response characteristics compared to both the PD controller and the FLC across varying trajectories. In reference [26], a novel fuzzy proportional-integral-derivative (PID)-type iterative learning control (ILC) approach was proposed to improve the performance of quadrotor UAVs. The approach combines the robustness of ILC, which is known for its ability to handle disturbances and system uncertainties, with fuzzy control to enhance system adaptability. The proposed method introduces a new control law based on the PID-ILC algorithm, specifically designed to address the chattering issue that typically arises when external disturbances affect ILC alone. Furthermore, fuzzy control was applied to adjust the PID parameters of the learning gain matrices, effectively mitigating the impact of uncertain factors on the system and enhancing control precision. Stability analysis using Lyapunov's theory confirmed the system's stability under the proposed control scheme. Simulation results, conducted using the Gazebo environment, demonstrated that the proposed ILC design is effective in improving the trajectory tracking performance of quadrotor UAVs. In reference [27], a fuzzy proportional-integral-derivative (FPID) controller is proposed to stabilize and track the desired trajectory of a quadcopter UAV. The controller is applied to both the inner and outer loops to stabilize the UAV's position and attitude. It adaptively adjusts the system's inputs based on tracking errors, offering improved performance over conventional PID controllers. Simulation results, obtained using MATLAB/Simulink, demonstrate that the fuzzy PID controller reduces position and attitude errors by 87% and 70%, respectively, compared to the traditional PID controller. In reference [28], a fuzzy self-tuning PID controller is proposed to control the altitude of a quadrotor. While traditional PID controllers offer several advantages, they require manual tuning of the key gains-proportional, integral, and derivative-making them less flexible. To address this limitation, the paper introduces a fuzzy self-tuning mechanism that automatically adjusts the PID gains for improved performance. The authors develop and compare

two control techniques: a standard linear PID controller and the proposed fuzzy self-tuning PID controller. In reference [29], the authors address the challenges of maintaining UAV stability during autonomous maneuvers, with a focus on attitude and position control. While proportional-integral-derivative (PID) controllers are commonly employed due to their structural simplicity and efficiency, they are limited by their fixed parameters, which may not suit varying mission requirements. To enhance performance, the study proposes a novel fuzzy-gain scheduling mechanism for PID controllers, designed to stabilize both position and altitude effectively. The proposed approach, implemented using the Robot Operating System (ROS) and integrated with the flight control unit, demonstrates robustness against uncertainties and external disturbances. Experimental results reveal that the method outperforms conventional PID controllers, achieving superior trajectory tracking performance and maintaining a position control error reduction of 2% under conditions with higher altitude errors. Recent advancements in quadrotor UAV control systems have introduced intelligent control techniques, such as fuzzy PID (FPID) and variable universe fuzzy PID (VF-PID) controllers. These methods, when combined with traditional PID control, significantly improve dynamic performance, stability, and robustness. Simulation studies have demonstrated that VF-PID controllers achieve reduced overshoot, faster response times, and enhanced robustness compared to conventional approaches [30].

In summary, efficient reference tracking for a quadrotor UAV using a PID controller is contingent on accurately tuning the PID parameter gains to enhance controller performance. However, traditional PID controllers are often inadequate for handling the nonlinear dynamics of quadrotor systems. To address this limitation, this study proposes a trajectory-tracking control method based on a Genetic Algorithm (GA)-Optimized Fuzzy PID controller tailored for nonlinear quadrotors. The method incorporates real-time adjustment of PID parameter gains using fuzzy logic, enabling precise trajectory control. The GA dynamically optimizes the Fuzzy PID parameters online, improving system performance in response to environmental variations. This approach overcomes the limitations of conventional fuzzy and fuzzy PID controllers, ensuring safe, stable, and accurate UAV trajectory tracking through adaptive parameter optimization.

B. CONTRIBUTIONS

The main contributions of this paper are summarized as follows:

- 1) **Comprehensive Analysis of Intelligent Control Strategies:** This study provides a detailed comparative analysis of three intelligent control strategies for UAV trajectory tracking-Fuzzy Logic Controller (FLC), Fuzzy PID (FPID) Controller, and Genetic Algorithm (GA)-Optimized Fuzzy PID (GA-FPID) Controller. The analysis highlights their respective strengths and weaknesses in addressing UAV control challenges.

- 2) **Optimization of Fuzzy PID Gains Using Genetic Algorithms:** The paper introduces the integration of Genetic Algorithms (GA) to optimize the gains of the fuzzy PID controller. This optimization enhances the controller's performance, achieving improved accuracy, robustness, and adaptability under dynamic operational conditions.
- 3) **Enhanced Methodological Contribution Beyond Basic Fuzzy Control:** Unlike studies that focus solely on fuzzy control, this research emphasizes the importance of combining heuristic optimization methods like GA with fuzzy control to achieve superior performance.
- 4) **Extensive Simulations and Rigorous Performance Evaluation:** Extensive simulations are conducted to evaluate the performance of all three control strategies across various operational scenarios, including uncertainties and external disturbances. These evaluations provide critical insights into the robustness and adaptability of each strategy.
- 5) **Demonstration of Superior Control Design:** The GA-Optimized Fuzzy PID Controller is shown to outperform conventional Fuzzy Logic and Fuzzy PID Controllers in terms of trajectory tracking precision, robustness against external disturbances, and adaptability to dynamic conditions.

This paper presents a comparative study of three control strategies for quadrotor stabilization: Fuzzy Logic Controller (FLC), Fuzzy PID Controller (FPID), and Genetic Algorithm-Optimized Fuzzy PID Controller (GA-FPID). The objective of this study is to evaluate the performance of the proposed methods in terms of integral time absolute error (ITAE) criteria, and trajectory tracking accuracy under varying conditions. By examining these approaches, the study aims to provide insights into the effectiveness of hybrid and optimization-based control strategies for quadrotor systems.

C. METHODOLOGY

In this study the methods used to address the control challenges associated with UAVs using intelligent control techniques were incorporated. The key steps of the methods are as follows:

Development of Control Models: Design and implement an FLC, a Fuzzy PID Controller, and a GA-Optimized Fuzzy PID Controller for UAV dynamics. Use a detailed nonlinear UAV model to evaluate their performance in virtual realistic scenarios (MATLAB environment).

Optimization Using Genetic Algorithms: Apply GA to optimize the fuzzy rules, membership functions, and PID parameters in the Fuzzy PID Controller. Ensure the optimized controller can adapt dynamically to variations in UAV dynamics and external disturbances.

Performance Evaluation: Test the controllers under multiple conditions, including: Trajectory Tracking: Following predefined flight paths. Disturbance Rejection: Handling wind gusts or sensor noise. Parameter Variations: Adapting

to changes in UAV weight or motor efficiency. Assess performance based on metrics such as error minimization (ITAE), response time, and energy efficiency.

Comparative Analysis: Identify strengths and weaknesses of each controller based on simulation results. Highlight the advantages of using GA optimization in fuzzy-based control for UAV applications. Expected Outcomes Demonstration of the limitations of traditional FLCs and Fuzzy PID Controllers in handling nonlinear UAV dynamics. Validation of the GA-Optimized Fuzzy PID Controller as a superior solution for UAV control, offering: Faster response and recovery. Reduced steady-state error and overshoot. Greater resilience to disturbances and parameter changes. Recommendations for deploying optimized intelligent control strategies in real-world UAV applications.

The remainder of this paper is organized as follows: Section II outlines the modeling of the quadrotor UAV, Section III presents the problem statement and the proposed solution, Section IV discusses the design of the controllers and the optimization algorithm, Section V provides the simulation results, while Section VI addresses the disturbance rejection and parameter variation handling capabilities, Section VII compares the controllers based on performance criteria derived from the cost function, Finally, Section VIII concludes the paper by summarizing the key findings and suggesting directions for future research.

II. SYSTEM MODELING

The modeling of a quadrotor system is a critical step in understanding its behavior and developing effective control strategies. Quadrotors, classified as unmanned aerial vehicles (UAVs) with four rotors, exhibit complex dynamics arising from the interaction of aerodynamics, mechanics, and control systems. This section outlines the fundamental concepts necessary for the system modeling of a quadrotor. This paper investigates the trajectory tracking control challenges associated with the 'X'(cross) configuration quadrotor UAV. The propulsion system of this UAV consists of four motors, as illustrated in Figure 1, where adjustments in rotor speeds are utilized to control both attitude and position [31].

Proposed Model: This study adopts the 'X' (cross) configuration for the quadrotor model. The primary advantage of this configuration lies in its superior stability compared to the plus configuration, as the cross configuration utilizes two rotors to stabilize and guide the quadrotor's movements [32], [34].

The subsequent section focuses on Dynamics and Kinematics of the quadrotor system. This analysis considers variables such as rotor thrust, torque, and the overall motion of the system in three-dimensional space. In the realm of the Rigid Body Control Model, we introduce how the quadrotor responds to external inputs and disturbances. This involves a comprehensive understanding of the forces and moments acting on the vehicle system. The subsequent segment encompasses the presentation of mathematical equations dictating the quadrotor's motion. This includes translational

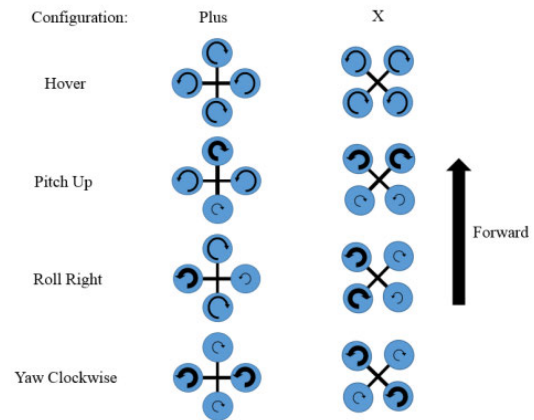


FIGURE 1. Quadrotor configurations and motion relative to propeller speed [31].

and rotational dynamics, along with the presentation of equations of motion and its derivations from reference frame in Figure 2. To precisely depict the position and attitude of the quadrotor UAV, the Earth-fixed coordinate system is utilized as (o^E, x^E, y^E, z^E) and the body coordinate system (o^B, x^B, y^B, z^B) are introduced.

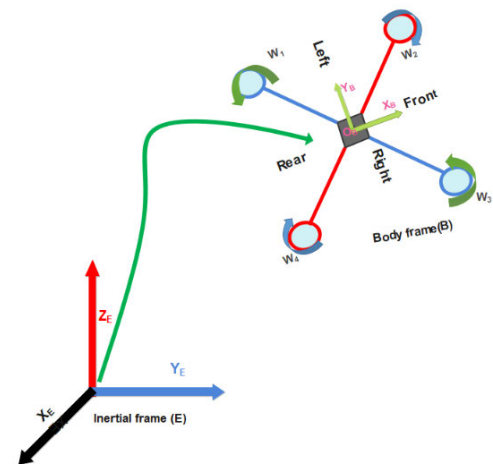


FIGURE 2. Reference frame [35].

The definition of the body structure and the coordinate system is shown in figure 2. Modeling the quadrotor UAV is complex, to simplify the model establishment, the following assumptions are considered [32], [34], [35].

- 1) **Rigid and symmetrical structure:** This assumption implies that the quadrotor has a fixed shape and its mass is uniformly distributed.
- 2) **Center of gravity:** Assuming that the quadrotor's center of gravity coincides with the body-fixed frame origin simplifies the analysis of the system dynamics. This assumption eliminates the need to consider the translation of the center of gravity.
- 3) **Rigid propellers:** Assuming that the propellers are rigid simplifies the modeling of the forces and moments generated by the propellers.

- 4) **Thrust and drag force:** The assumption that the thrust and drag forces are proportional to the square of the propeller's speed.
- 5) **Motor dynamics:** Neglecting the motor's resistance and inductance implies that the motor dynamics are significantly faster than the dynamics of the quadrotor itself. This assumption simplifies the modeling process by assuming that the motor instantaneously responds to changes in voltage and produces the corresponding output speed.

The orientation of the quadrotor is given by the rotation matrix R which depends on the three Euler angles (ϕ, θ, ψ) .

$$R(x, \phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \quad (1)$$

$$R(y, \theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \quad (2)$$

$$R(z, \psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix} = {}^E_B R \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad (4)$$

Transfer Matrix(T): $A^E = [\phi, \theta, \psi]^T$ is in Inertial frame and Angular Velocity $(\omega^B = [p, q, r]^T)$ is in Body frame. To relate the two, Transfer matrix(T) is used as [32], [34], and [35]:

$$\dot{A}^E = T \omega^B \quad (5)$$

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = T \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (6)$$

$$T = \begin{bmatrix} 1 & \sin(\phi) \tan(\theta) & \cos(\phi) \tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) \sec(\theta) & \cos(\phi) \sec(\theta) \end{bmatrix} \quad (7)$$

$$T^{-1} = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\phi) & \sin(\phi) \cos(\theta) \\ 0 & -\sin(\phi) & \cos(\phi) \cos(\theta) \end{bmatrix} \quad (8)$$

Newton-Euler Method Formalism: The flight dynamics of the quadrotor UAV will be modeled using the Newton-Euler method [32], [34], [35], [36]. The equations of motion, which combine the translational and rotational parameters which is described in Table 1 of the 6DOF system (consisting of 3 position and 3 angular orientation variables), are derived based on Euler's two laws of motion-Newton's second law and Newton's first law (also known as the law of inertia) [37]. The formulation commonly involves the following parameters [32], [34].

- Linear Velocity $(V^B = [u, v, w]^T)$

- Angular Velocity $(\omega^B = [p, q, r]^T)$
- Orientation Vector $(A^E = [\phi, \theta, \psi]^T)$
- Position Vector $(L^E = [x, y, z]^T)$

$$\begin{bmatrix} \text{Roll rate} \\ \text{Pitch rate} \\ \text{Yaw rate} \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_B \quad (9)$$

External Forces And Moments:

A. FORCE

The force exerted on a quadcopter influences its translational motion [38]. According to Newton's second law, this force is expressed as:

$$\sum F = F_{th} + F_{grav} = m\vec{a} \quad (10)$$

In the body-fixed reference frame, translational dynamics are influenced by two main forces:

1) THRUST FORCE

The thrust force is the combined upward thrust generated by the propellers. This force is quadratically related to the angular velocities of the rotors as $F_i = b\omega_i^2$ [32], [38]. The total thrust in the body frame is given by:

$$F_{th}^B = F_1 + F_2 + F_3 + F_4 = b(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \quad (11)$$

In the Earth's inertial reference frame, the thrust force can be expressed using the rotation matrix:

$$F_{th}^E = {}^E_B R \cdot F_{th}^B \quad (12)$$

where

$$F_{th}^E = \begin{bmatrix} (\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi)) F_{th}^B \\ (\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi)) F_{th}^B \\ (\cos(\phi) \cos(\theta)) F_{th}^B \end{bmatrix} \quad (13)$$

2) GRAVITATIONAL FORCE

The gravitational force, in the Earth's inertial reference frame, is expressed as:

$$F_{grav}^E = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} \quad (14)$$

3) NET FORCE

In the Earth's inertial frame, the net force is given by:

$$\sum F^E = F_{th}^E + F_{grav}^E = m\vec{a} \quad (15)$$

Let:

$$U_1 = F_{th}^B = b(\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2)$$

From the above, the translational dynamics in the Earth's inertial reference frame are:

$$\begin{cases} \ddot{x} = (\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi)) \frac{U_1}{m} \\ \ddot{y} = (\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi)) \frac{U_1}{m} \\ \ddot{z} = (\cos(\phi) \cos(\theta)) \frac{U_1}{m} - g \end{cases} \quad (16)$$

B. ROTATIONAL DYNAMICS

The rotational dynamics, accounting for net torques and reaction forces, are described as:

$$\sum \tau = M_{\text{thrust}} + F_{\text{gyro}} \quad (17)$$

These dynamics can be determined using both the body-fixed and Earth inertial frames.

1) MOMENTS OF THRUST FORCES

For this paper, we assume that clockwise (CW) propeller rotation is positive, and counterclockwise (CCW) rotation is negative [38]. The moments of thrust forces in the body frame are given by:

$$\begin{cases} \tau_x^B = \frac{\sqrt{2}}{2} lb (\omega_1^2 + \omega_2^2 - \omega_3^2 - \omega_4^2) \\ \tau_y^B = \frac{\sqrt{2}}{2} lb (\omega_1^2 - \omega_2^2 - \omega_3^2 + \omega_4^2) \\ \tau_z^B = -d\omega_1^2 + d\omega_2^2 - d\omega_3^2 + d\omega_4^2 \end{cases} \quad (18)$$

From Equation (18), let $\tau_x^B = U_2$ (roll), $\tau_y^B = U_3$ (pitch), and $\tau_z^B = U_4$ (yaw). Balance is achieved when the total rotor speeds sum to zero, ensuring stability, as propellers 2 and 4 rotate CW and propellers 1 and 3 rotate CCW. Otherwise, imbalance causes gyroscopic effects that influence roll and pitch rates.

The gyroscopic force in the body frame is:

$$F_{\text{gyro}}^B = J_r \begin{bmatrix} q w_r \\ -p w_r \\ 0 \end{bmatrix} \quad (19)$$

where $w_r = \omega_1 - \omega_2 + \omega_3 - \omega_4$ and J is the moment of inertia.

In the inertial frame, the rate of change of angular momentum $\vec{L}$ equals the total torques acting on the quadrotor. Since the geometry is symmetric, the inertia tensor is represented as:

$$J = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

The total torque in the body frame is:

$$\sum M^B = M_{\text{thrust}}^B + F_{\text{gyro}}^B = \dot{\vec{L}} + \vec{\omega} \times \vec{L} \quad (20)$$

The angular acceleration in the body frame is expressed as:

$$\begin{cases} \dot{p} = \frac{U_2}{I_{xx}} - \frac{qr(I_{yy} - I_{zz})}{I_{xx}} + \frac{Jr w_r q}{I_{xx}} \\ \dot{q} = \frac{U_3}{I_{yy}} - \frac{rp(I_{zz} - I_{xx})}{I_{yy}} - \frac{Jr w_r p}{I_{yy}} \\ \dot{r} = \frac{U_4}{I_{zz}} - \frac{pq(I_{xx} - I_{yy})}{I_{zz}} \end{cases} \quad (21)$$

The quadrotor's dynamics are described by four control inputs, represented as:

$$\mathbf{U} = [U_1 \quad U_2 \quad U_3 \quad U_4]^T,$$

where:

- U_1 represents the total thrust force (throttle) applied to the quadrotor body,
- U_2 is the torque affecting the roll angle,
- U_3 is the torque affecting the pitch angle, and
- U_4 is the torque affecting the yaw (heading) angle.

The quadrotor's translational and rotational dynamics are governed by the following equations [32], [38]:

$$\begin{cases} \ddot{x} = (\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi)) \frac{U_1}{m} \\ \ddot{y} = (\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi)) \frac{U_1}{m} \\ \ddot{z} = (\cos(\phi) \cos(\theta)) \frac{U_1}{m} - g \\ \ddot{\phi} = \frac{U_2}{I_{xx}} - \frac{(I_{xx} - I_{yy})}{I_{xx}} \dot{\theta} \dot{\psi} + \frac{J_r \dot{\theta} w_r}{I_{xx}} \\ \ddot{\theta} = \frac{U_3}{I_{yy}} - \frac{(I_{xx} - I_{zz})}{I_{yy}} \dot{\phi} \dot{\psi} - \frac{J_r \dot{\phi} w_r}{I_{yy}} \\ \ddot{\psi} = \frac{U_4}{I_{zz}} - \frac{(I_{yy} - I_{xx})}{I_{zz}} \dot{\phi} \dot{\theta} \end{cases} \quad (22)$$

The desired roll (ϕ_r) and pitch (θ_r) angles are derived from Equation (22) as:

$$\begin{cases} \phi_r = \tan^{-1} \left(\frac{\cos(\theta_r)}{\ddot{z} + g} (\ddot{x} \sin(\psi) - \ddot{y} \cos(\psi)) \right) \\ \theta_r = \tan^{-1} \left(\frac{1}{\ddot{z} + g} (\ddot{x} \cos(\psi) + \ddot{y} \sin(\psi)) \right) \end{cases} \quad (23)$$

The control inputs are used to calculate the rotor angular velocities, with their inverse relationship given as:

$$\begin{cases} \omega_1 = \sqrt{\frac{U_1}{4b} + \sqrt{2} \frac{U_2}{4bl} + \sqrt{2} \frac{U_3}{4bl} + \frac{U_4}{4d}} \\ \omega_2 = \sqrt{\frac{U_1}{4b} + \sqrt{2} \frac{U_2}{4bl} - \sqrt{2} \frac{U_3}{4bl} - \frac{U_4}{4d}} \\ \omega_3 = \sqrt{\frac{U_1}{4b} - \sqrt{2} \frac{U_2}{4bl} - \sqrt{2} \frac{U_3}{4bl} + \frac{U_4}{4d}} \\ \omega_4 = \sqrt{\frac{U_1}{4b} - \sqrt{2} \frac{U_2}{4bl} + \sqrt{2} \frac{U_3}{4bl} - \frac{U_4}{4d}} \end{cases} \quad (24)$$

TABLE 1. Parameters description [32].

Parameters				Description	Units
x	y	z		Linear position vector	m
ϕ	θ	ψ		Angular position vector	rad
u	v	w		Linear Velocity vector	m/s
p	q	r		Angular velocity vector	rad/s
I_{xx}	I_{yy}	I_{zz}		Moment of inertia vector	kg · m ²
F_{th}				Total thrust generated by rotors	N
τ_x	τ_y	τ_z		Control torques	N.m
g				Gravitational force	m/s ²
m				Total mass	kg
ω_1	ω_2	ω_3	ω_4	Rotors speeds vector	rad/s
b				Thrust coefficient	N.s s ²
l				Motor to center length	m
w_r				residual rotor speed	rad/s
d				Drag coefficient	N.m.s.s
J_r				rotor inertia vector	kg · m ²

C. VIRTUAL CONTROL

The quadrotor is under-actuated and which is 4 control input and 6 states to be controlled. This virtual control is not used in practical applications it is used for simulation environments only this is its limitation. Therefore, the states are indirectly controlled from translation dynamics and as [32], [34], [35], and [38]:

$$\begin{cases} V_x = \ddot{x} = (\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi)) \frac{U_1}{m} \\ V_y = \ddot{y} = (\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi)) \frac{U_1}{m} \\ V_z = \ddot{z} = (\cos(\phi) \cos(\theta)) \frac{U_1}{m} - g \end{cases} \quad (25)$$

where V_x, V_y, V_z are virtual control inputs. Following a straightforward calculation, the lift force or altitude control signal, along with the desired Euler angles, is expressed as shown in Equation (26).

$$\begin{cases} U_1 = m\sqrt{V_x^2 + V_y^2 + (V_z + g)^2} \\ \theta_r = \arctan\left(\frac{V_x \cos \psi_r + V_y \sin \psi_r}{V_z + g}\right) \\ \phi_r = \arctan\left(\frac{\cos \theta_r (V_x \sin \psi_r - V_y \cos \psi_r)}{V_z + g}\right) \end{cases} \quad (26)$$

III. PROBLEM STATEMENT

Unmanned Aerial Vehicles (UAVs) are increasingly relied upon in diverse fields such as surveillance, disaster management, and logistics due to their flexibility and autonomy. However, their inherently nonlinear dynamics, sensitivity to environmental disturbances, and complex control requirements pose significant challenges for achieving stable, precise, and efficient flight control. Traditional control methods, such as Proportional-Integral-Derivative (PID) controllers, often struggle to address these challenges effectively in dynamic and uncertain conditions.

To overcome these limitations, intelligent control strategies, such as Fuzzy Logic Controllers (FLC) and their

advanced variants, have emerged. While FLCs offer adaptability and robustness, they often require manual tuning and may lack optimal performance in complex scenarios. The integration of optimization techniques, such as Genetic Algorithms (GA), holds the potential to enhance the capabilities of fuzzy-based controllers. Despite these advancements, a comprehensive comparative analysis of these methods in the context of UAV control has not been thoroughly explored, particularly in terms of handling nonlinearities, disturbances, and real-time adaptability.

A. PROPOSED SOLUTION

This study proposes a GA-optimized fuzzy PID controller (GAFPID) to address the control challenges associated with UAVs.

IV. CONTROLLER DESIGN

In the following section the controllers FLC, FPID, and GAFPID is designed. A Fuzzy Logic Controller (FLC) uses “fuzzy” rules based on human-like reasoning to control systems, handling uncertainties and complexities better than traditional controllers (PID). Fuzzy PID controller combines the structure of a standard PID controller with fuzzy logic to adapt its gains based on system conditions online to improve performance in nonlinear dynamic systems whereas a GA optimized Fuzzy PID (GAFPID) controller further enhances this by using a genetic algorithm to automatically fine-tune the fuzzy logic parameters for optimal control of dynamic nonlinear plants like quadrotor systems.

A. FUZZY CONTROLLER DESIGN

The fuzzy logic controller (FLC) provides a means of converting a linguistic control strategy [5]. The heuristic method consists of modeling and understanding in accordance with previous experience, rules-of-thumb and often-used strategies. A heuristic rule is a logical implication of the form: If <condition> Then <consequence>, or in a typical control situation: If <condition> Then <action>. Rules associate conclusions with conditions. Therefore, the heuristic method is actually similar to the experimental method of constructing a table of inputs and corresponding output values where instead of having crisp numeric values of input and output variables [39]. The fuzzy control goals are to reach the best transient and steady-state performances. To obtain these performances, the rule base and fuzzy set membership functions of the fuzzy logic controllers must be adjusted [40]. Two approaches are developed, One approach is to use an operator to change the fuzzy controller parameters by trials and errors another approach is to design a self organizer fuzzy controller for temperature control were used in [40]. Membership functions and rule base For the FLC's related to the positions, the error is normalized to the interval $[-1 \ 1]$, the error rate is confined to the range $[-3 \ 3]$ and the output is also normalized to the range $[-1 \ 1]$ similarly for attitude the error is normalized to the interval $[-1 \ 1]$, the

TABLE 2. Fuzzy logic control rule base [41].

$de \backslash e$	Nhigh	Nlow	Zero	Plow	Pfast
Nfast	Zero	Psmall	Psmall	Pbig	Pbig
Nslow	Nsmall	Zero	Psmall	Psmall	Pbig
Zero	Nbig	Nsmall	Zero	Psmall	Pbig
Pslow	Nbig	Nsmall	Nsmall	Zero	Psmall
Pfast	Nbig	Nbig	Nbig	Nsmall	Zero

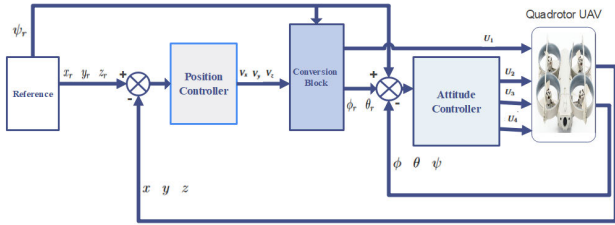


FIGURE 3. Control scheme block diagram [32].

error rate is confined to the range $[-2 \ 2]$ and the output is also normalized to the range $[-1 \ 1]$ [41].

Rules are developed as if-then statements based on heuristics and experience. For altitude control, IF error z is “Nhigh” and errordot z is “Pfast” THEN PWM is “Nbig”. The heuristic interpretation of this rule is as follows: if actual altitude is much higher than desired, thus error is negative high, and the error rate i.e velocity in Z direction is positive fast meaning a high upward velocity, then the output is negative big, i.e. a sharp decrease in PWM value for each motor thereby causing quick deceleration. A total of 25 rules have been developed for each FLC [41]. MATLAB’s Fuzzy Logic toolbox was used to develop the controllers, which use the Mamdani-type inference method and the centroid method for defuzzification and the complete rule base is presented in the Table 2.

To enhance the tracking task, the deviation control quantity at each moment is correlated with the entire past state, generalized control scheme block diagram is in figure 3.

$$\begin{cases} e(t) = r(t) - y(t) \\ de(t) = \frac{de(t)}{dt} \end{cases} \quad (27)$$

where, $r = [\phi_r \ \theta_r \ \psi_r \ x_r \ y_r \ z_r]^T$ denotes the desired reference and $y = [\phi \ \theta \ \psi \ x \ y \ z]^T$ represents the output of the controlled quadrotor.

B. FUZZY PID CONTROLLER DESIGN

The approach proposed in this paper is to use a fuzzy system for online tuning of PID gains [42] in the control algorithm of a quadrotor UAV as the block diagram shows in figure 11. This is necessary because the quadrotor is highly nonlinear and underactuated, making classical PID control inadequate.

The first step in this approach is to establish a set of tuning rules or fuzzy IF-THEN rules for the PID gains. These rules are designed based on the specific requirements and characteristics of the quadrotor system. The rules determine how the PID gains should be adjusted in response to various inputs and conditions.

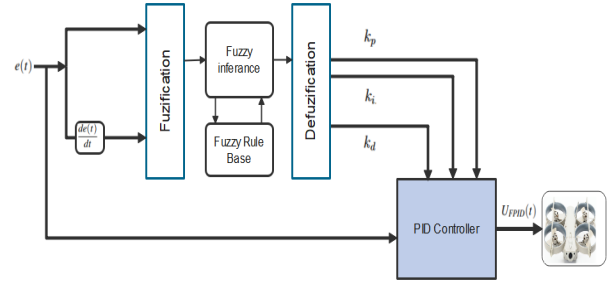


FIGURE 4. Structure of fuzzy PID controller [32].

TABLE 3. Fuzzy rules for K_d [42].

$de \backslash e$	NB	NM	NS	Z	PS	PM	PB
NB	M	B	VB	VVB	VB	B	M
NM	S	M	B	VB	B	M	S
NS	VS	S	M	B	M	S	VS
Z	VVS	VS	S	M	S	VS	VVS
PS	VS	S	M	B	M	S	VS
PM	S	M	B	VB	B	M	S
PB	M	B	VB	VVB	VB	B	M

TABLE 4. Fuzzy rules for K_p and K_i [42].

$de \backslash e$	NB	NM	NS	Z	PS	PM	PB
NB	M	S	VS	VVS	VS	S	M
NM	B	M	S	VS	S	M	B
NS	VB	B	M	S	M	B	VB
Zo	VVB	VB	B	M	B	VB	VVB
PS	VB	B	M	S	M	B	VB
PM	B	M	S	VS	S	M	B
PB	M	S	VS	VVS	VS	S	M

Once the tuning rules are defined, they are combined into a fuzzy system that is capable of adjusting the PID gains on-the-fly. This fuzzy system utilizes the inputs from the quadrotor’s feedback signals to determine the appropriate changes in the PID gains for better control performance. By implementing the fuzzy system for online PID parameter gain tuning, the control algorithm of the quadrotor can optimize its performance in fly, accommodating for the system’s nonlinearity and underactuated nature [42].

A set of linguistic rules were used in the fuzzy reasoning inference(sets) block to get the parameters K_p , K_i , K_d automatically [42]. Seven fuzzy labels (Negative Big (NB), Negative Medium(NM), Negative Small (NS), Zero (ZO), Positive Small (PS), Positive Medium (PM)and Positive Big (PB)) are used for the fuzzy input variables and seven fuzzy labels (very very small(VVS), very small(VS), Small (S), Medium (M), Big(B), Very big(VB) and very very big(VVB))for the fuzzy output variable. These linguistic variables are listed in Tables 3 and 4 which contain the 49 rules for the Fuzzy PID controller. The membership functions for the input variables are defined with the triangular and the output is Gaussian [43], [44].

For the Fuzzy logic control related to attitude and position, the error is normalized to the interval $[-1 \ 1]$ in figure 5, the error rate is confined within the range $[-10 \ 10]$ in

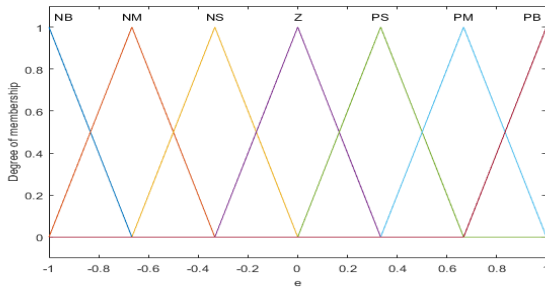


FIGURE 5. Fuzzy membership function for error [32].

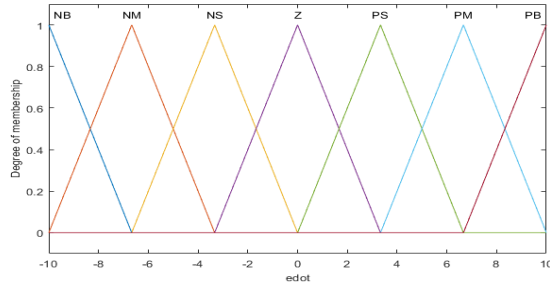


FIGURE 6. Fuzzy membership function for the derivative of error [32].

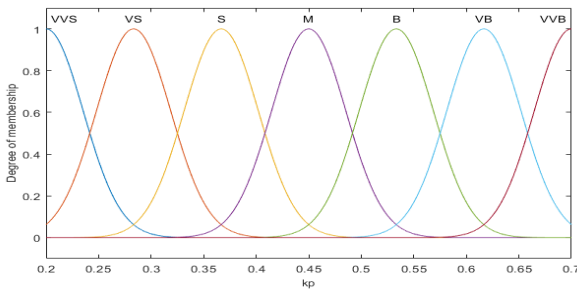
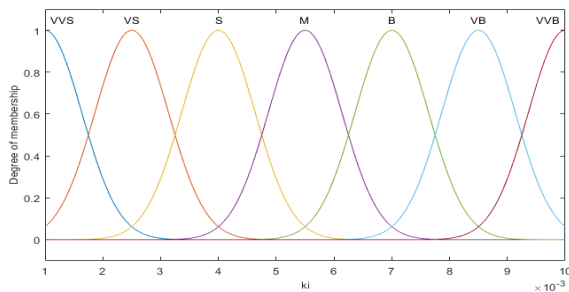
FIGURE 7. Fuzzy membership function for the output K_p [32].FIGURE 8. Fuzzy membership function for the output K_i [32].

figure 6 and the output is also normalized to the ranges $[0.2 \ 0.7]$, $[0.001 \ 0.01]$ and $[0.1 \ 0.15]$ for proportional, integral and derivative gains in figures 7, 8 and 9 respectively. For the quadcopter positions and attitude dynamic control requirement, the control system is identical after defining the rule base, the next step is to determine the membership functions (MFs) for each parameter. By employing the product-sum inference and center of gravity defuzzification method, the overall system outputs of each Fuzzy Logic Control can be obtained [42], [43], [44].

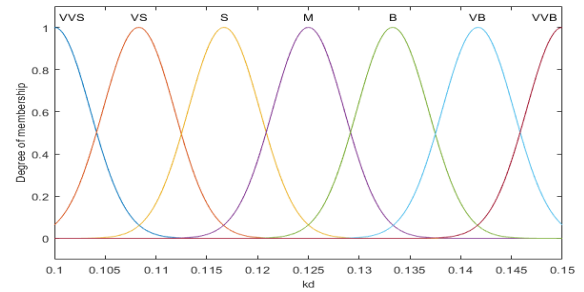
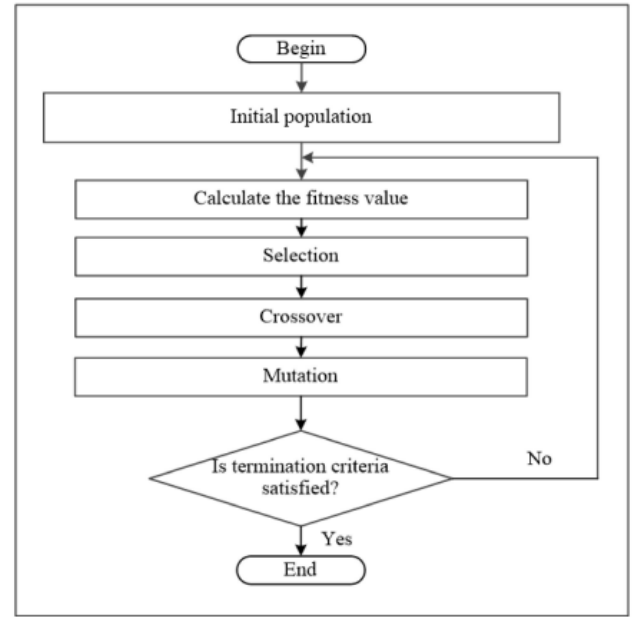
FIGURE 9. Fuzzy membership function for the output K_d [32].

FIGURE 10. Flowchart of the standard genetic algorithm (GA) [45].

fuzzy controller outputs can be formulated as:

$$U_{FPID}(t) = k_p e(t) + k_i \int_0^t e(t) dt + k_d \frac{de(t)}{dt} \quad (28)$$

C. GA-OPTIMIZED FUZZY PID CONTROLLER(GAFPID)

The Genetic Algorithm (GA)-Optimized Fuzzy PID Controller represents a hybrid control strategy that combines the adaptability of fuzzy logic with the precision of PID (Proportional-Integral-Derivative) control, enhanced by the optimization capabilities of a genetic algorithm. In this section, the previously designed Fuzzy PID (FPID) controller is further optimized to enhance the tracking accuracy of the quadrotor during trajectory tasks. A Genetic Algorithm (GA) is employed to optimize the scaling factors associated with the PID parameters in the FPID controller. The cost function used for optimization is the Integral Time Absolute Error (ITAE). The proposed control strategy is depicted in Figure 12.

1) GENETIC ALGORITHM OPTIMIZATION(GA)

Genetic Algorithm (GA) is an evolutionary optimization technique inspired by the principles of natural selection and

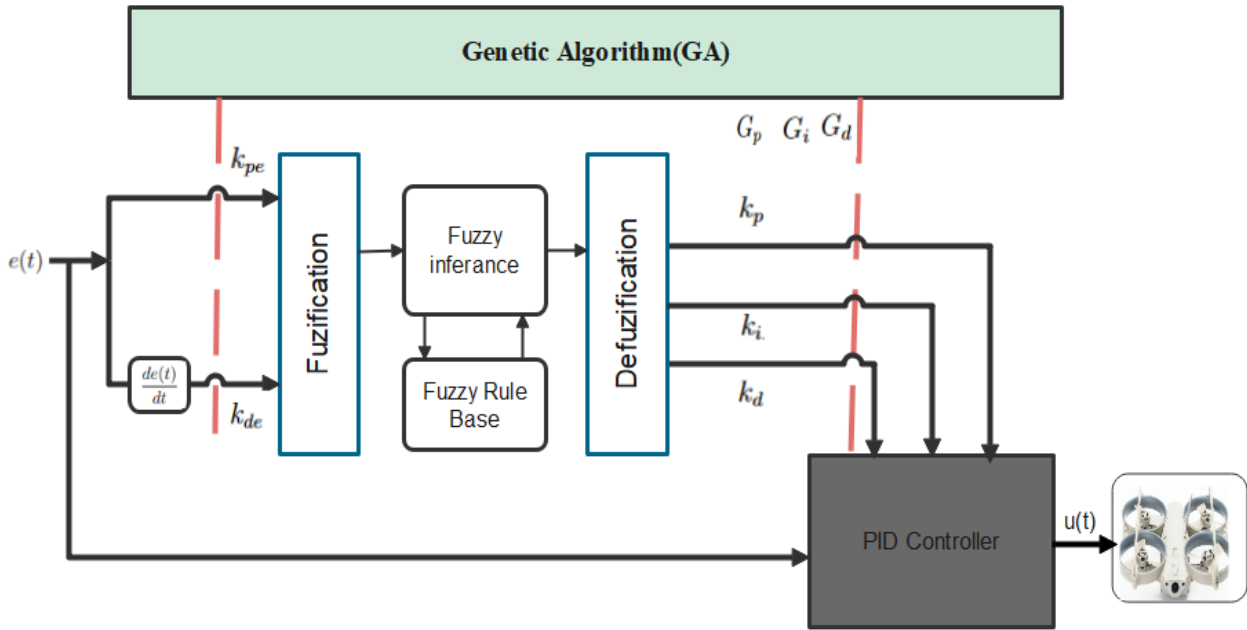


FIGURE 11. Structure of GA optimized fuzzy-PID controller [33].

Algorithm 1 Genetic Algorithm to Tune Scaling Parameters

Initialization Set the number of variables:

$no_var \leftarrow 30$

Define the lower bounds:

$lb \leftarrow [0, 0, \dots, 0]$ (30 zeroes)

Define the upper bounds:

$ub \leftarrow [200, 200, \dots, 200, 5, 5, \dots, 5]$

(15 values of 200 followed by 15 values of 5) **Set**

Genetic Algorithm Options Configure the GA parameters:

$ga_opt \leftarrow \{ 'Display' \leftarrow 'off', 'Generations' \leftarrow 100, 'PopulationSize' \leftarrow 100, 'PlotFcn' \leftarrow gaplotbestf \}$

Define Objective Function Define the objective function to evaluate the performance of scaling parameters k :

$obj_fn(k) \leftarrow optimization_PID(k)$

Run the Genetic Algorithm Call the GA solver with the defined parameters:

$[k, best] \leftarrow ga(obj_fn, no_var, lb, ub, ga_opt)$

Output Results Return the optimized scaling parameters k and the best objective value $best$.

Algorithm 2 Fitness Function for Genetic Algorithm

Input Receive scaling parameters k . **Assign Parameters** Assign k to the simulation environment:

$assign_to_base('k', k)$

Run Simulation Execute the simulation model:

$results \leftarrow simulate_model('nigatu.slx')$

Calculate Cost Extract the ITAE value from the simulation results:

$cost \leftarrow ITAE[end]$

Output Return the computed cost.

dimensionality, nonlinearity, or multiple local optima [45]. The flow chart of the standard Genetic algorithm is depicted in Figure10.

D. PERFORMANCE INDICES

In this paper listed performance indices in the equation (29) are mathematical measures used to evaluate the performance of control systems. Integral of Squared Error (ISE), Integral of Time Squared Error (ITSE), Integral of Absolute Error (IAE), and Integral of Time Absolute Error (ITAE) were used to analyze the performance of controllers [32]. ITAE penalizes long settling time and control errors. It aims to minimize the integral of the absolute value of errors over time. Systems designed using this criterion tend to have small overshoots and minimum damped oscillations. The cost functions(fitness functions) of the GA optimization

genetics. It is widely used to solve complex optimization problems where traditional methods may fail due to high

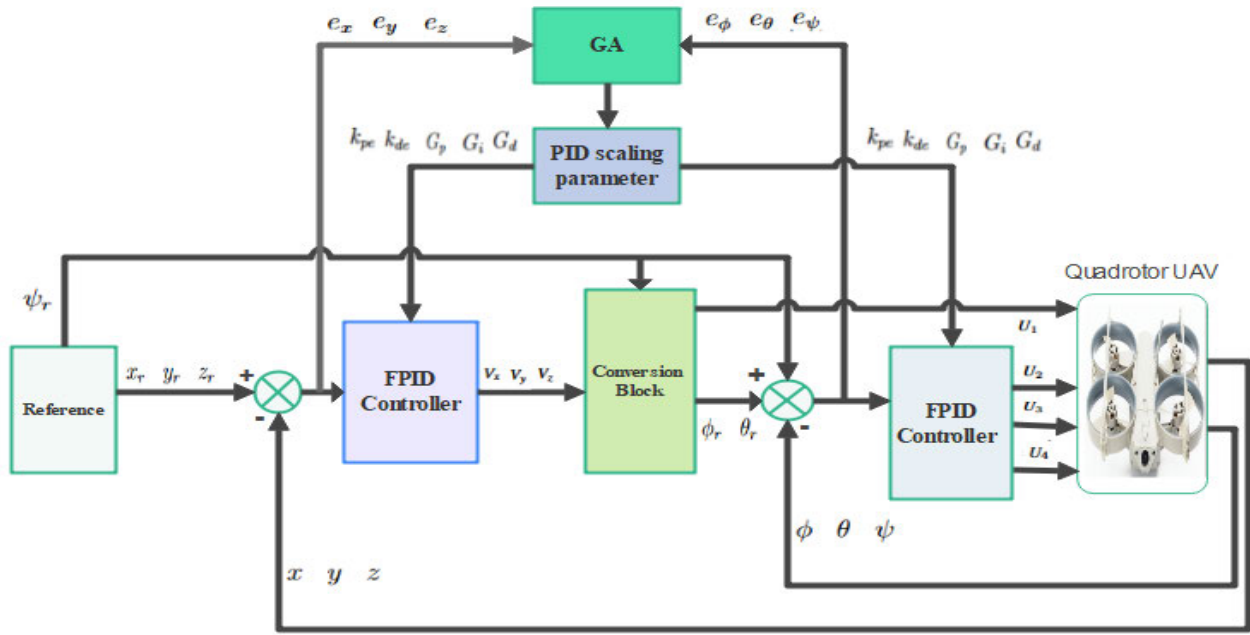


FIGURE 12. Proposed control scheme(GAFPID).

problem are chosen according to each controlled variable as the integral Time absolute error (ITAE) criteria, hence the weighted sum of all the six dynamics ITAE is used as a cost function to get optimal scaling parameters.

$$\begin{cases} ISE = \int_0^{\infty} e^2(t)dt \\ ITSE = \int_0^{\infty} te^2(t)dt \\ IAE = \int_0^{\infty} |e(t)|dt \\ ITAE = \int_0^{\infty} t|e(t)|dt \end{cases} \quad (29)$$

were, $e(t)$ is the tracking error and t is the time.

The output variables of the optimized fuzzy tuning gains k_p, k_i, k_d are obtained by using the fuzzy reasoning inference(fuzzy rules) and the scaling factors $k_{pe}, k_{de}, G_p, G_i, G_d$ are obtained using Genetic optimization algorithm (GA).

The proposed GA-optimized Fuzzy PID controller is in equation (30) as below form.

$$U_{GAFPID}(t) = G_p k_p(t) + G_i k_i \int_0^t e(t)dt + G_d k_d \frac{de(t)}{dt} \quad (30)$$

V. SIMULATION

For simulation the quadrotor parameters in Table 5 is incorporated in MATLAB working environment. To validate the efficacy and precision of the proposed quadrotor UAV controller for trajectory tracking, a comparison is made with the standard FLC controller and the fuzzy PID controller (FPID) across various scenarios with the proposed GAFPID controller. The proposed GAFPID controller is specifically

TABLE 5. Parameters values [32].

Parameter	Value
I_{xx}	8.5532×10^{-3}
I_{yy}	8.5532×10^{-3}
I_{zz}	1.476×10^{-2}
g	9.81×10^0
m	1×10^0
b	7.66×10^{-5}
d	5.63×10^{-6}
l	2.2×10^{-1}
J_r	0.1×10^{-3}

introduced for this purpose. Simulation results indicate that the quadrotor UAV, under the GAFPID controller, exhibits superior performance in completing trajectory-tracking tasks. To ensure the objectivity and fairness of the controller comparison, numerical simulations are conducted using the Simulink platform of MATLAB 2023a. The simulation results for helical and rectangular trajectories are in the below section.

For helical trajectory tracking the reference signal is:

$$\begin{cases} x_r(t) = \cos(1.05t) \\ y_r(t) = \sin(1.05t) \\ z_r(t) = 3 + (t/4) \\ \psi_r(t) = \pi/4 \end{cases} \quad (31)$$

The simulation results in undisturbed conditions for helical trajectory tracking are depicted in Figures 13,14,15, and 16.

The simulation results for rectangular trajectory tracking in undisturbed conditions are depicted in Figures 17,18,19, and 20.

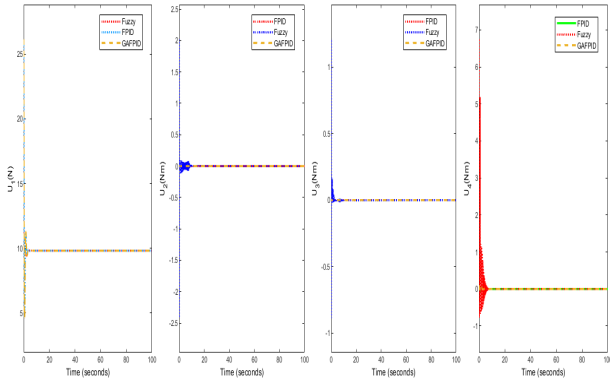


FIGURE 13. Control effort for helical trajectory tracking.

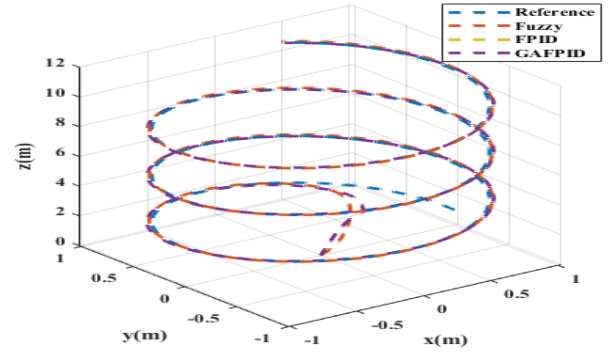


FIGURE 16. 3D Plot for helical trajectory tracking.

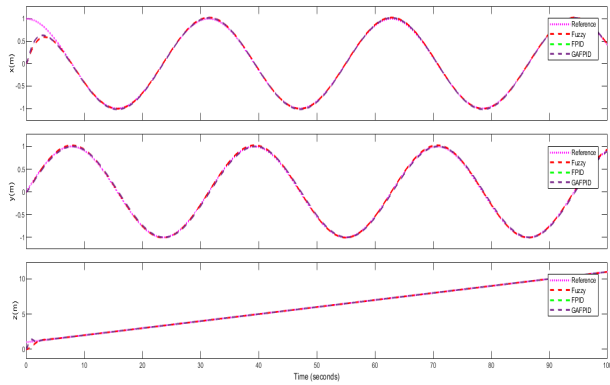


FIGURE 14. Position tracking response for helical trajectory tracking.

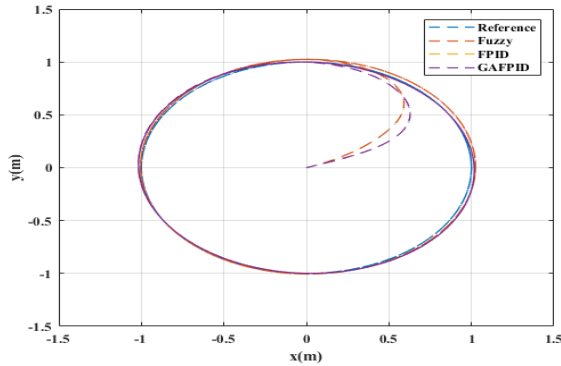


FIGURE 15. xy tracking response for helical trajectory tracking.

VI. DISTURBANCE REJECTION CAPABILITY AND ROBUSTNESS TO PARAMETER VARIATION

This section focuses on evaluating the disturbance rejection capability and robustness of the proposed GAFFPD controller under parameter variations. To simulate real-world conditions, a random force was introduced as an input disturbance to represent wind forces affecting the x , y , and z positions of the system. Additionally, to account for parameter variation, 25% of the total quadrotor mass was added to the system. In this study, both the input disturbance and parameter variation were applied simultaneously.

Algorithm 3 Pseudocode for Rectangular Trajectory

```

1: if  $t < 10$ 
2:    $x_r \leftarrow 0.5t$ 
3:    $y_r \leftarrow 0$ 
4:    $z_r \leftarrow 5$ 
5: else if  $10 \leq t \leq 20$ 
6:    $x_r \leftarrow 5$ 
7:    $y_r \leftarrow 0.5t - 5$ 
8:    $z_r \leftarrow 5$ 
9: else if  $20 < t \leq 30$ 
10:   $x_r \leftarrow -0.5t + 15$ 
11:   $y_r \leftarrow 5$ 
12:   $z_r \leftarrow 5$ 
13: else if  $30 < t \leq 40$ 
14:   $x_r \leftarrow 0$ 
15:   $y_r \leftarrow -0.5t + 20$ 
16:   $z_r \leftarrow 5$ 
17: else
18:   $x_r \leftarrow 0$ 
19:   $y_r \leftarrow 0$ 
20:   $z_r \leftarrow 5$ 
21: end if

```

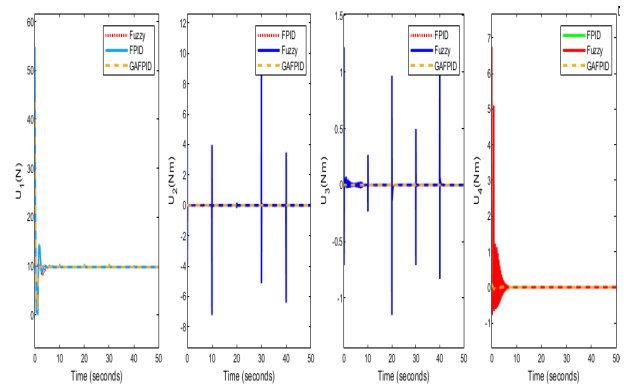


FIGURE 17. Control effort for rectangular trajectory tracking.

The disturbance force was set to 0.15 N in the x and z positions and 0.1 N in the y position. These forces were

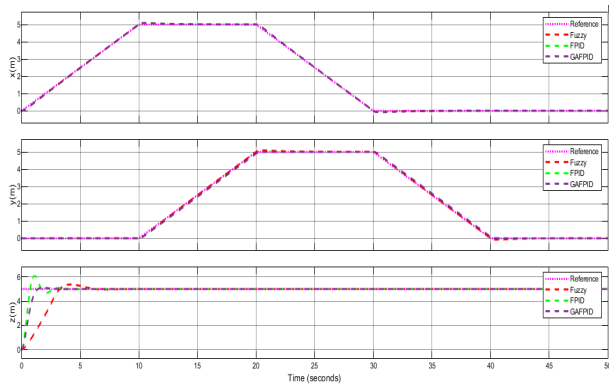


FIGURE 18. Position tracking response for rectangular trajectory tracking.

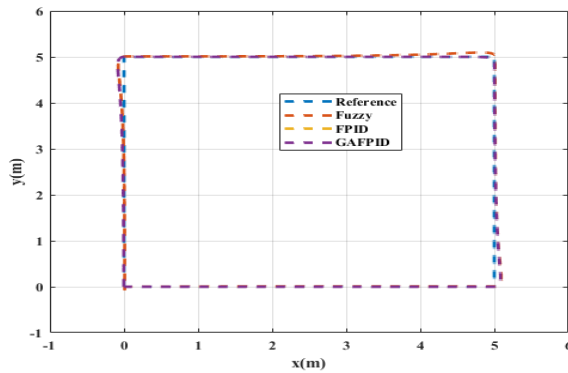


FIGURE 19. xy tracking response for rectangular trajectory tracking.

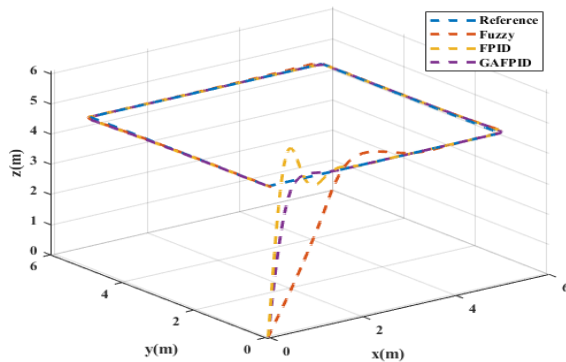


FIGURE 20. 3D plot for rectangular trajectory tracking.

applied during specific time intervals: from 5 to 10 seconds for the x position, from 10 to 15 seconds for the y position, and from 15 to 20 seconds for the z position.

The simulation results for rectangular trajectory, incorporating the effects of the input disturbances and parameter variations, are presented in Figures 21, 22, 23, and 24.

A. VALIDATION USING UAV TOOLBOX 3D ANIMATION

The closed-loop quadrotor system, integrated with the proposed GAFFPD controller, was validated using the UAV Toolbox 3D Animation, as shown in Figure 25. Real-time simulations in a virtual reality environment demonstrated the controller's ability to guide the system along desired trajectories. The Simulink 3D Animation package was

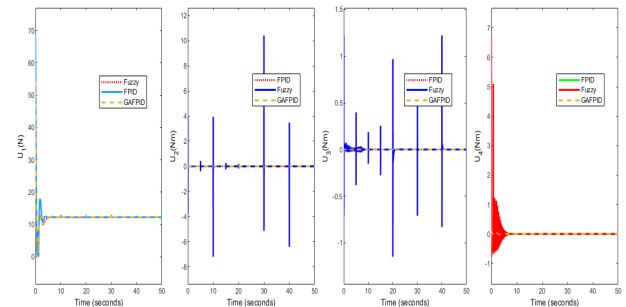


FIGURE 21. Control effort for rectangular trajectory tracking in the presence of input disturbances and parameter variation.

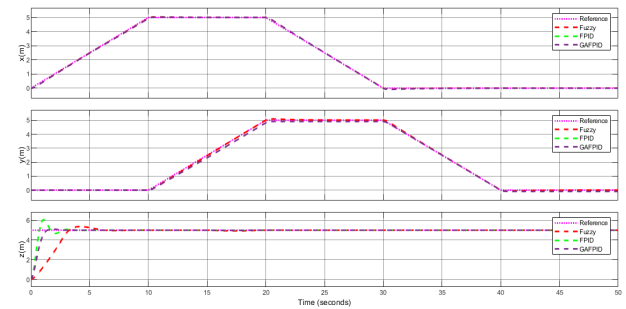


FIGURE 22. Position tracking response for rectangular trajectory tracking in the presence of input disturbances and parameter variation.

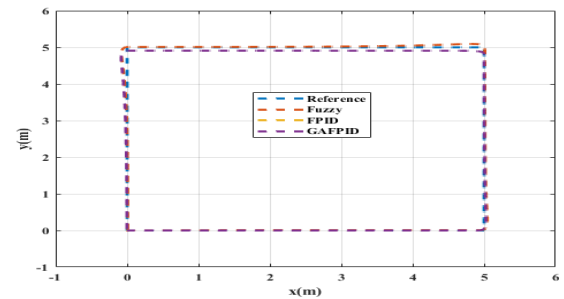


FIGURE 23. xy tracking response for rectangular trajectory tracking in the presence of input disturbances and parameter variation.

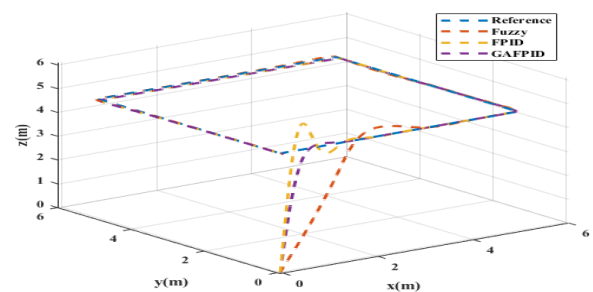


FIGURE 24. 3D plot for rectangular trajectory tracking in the presence of input disturbances and parameter variation.

employed to visualize and validate system dynamics. This package connects Simulink models and MATLAB algorithms to 3D graphical objects, providing a powerful tool for analyzing dynamic system behavior in a virtual reality setting using the Virtual Reality Modeling Language (VRML).

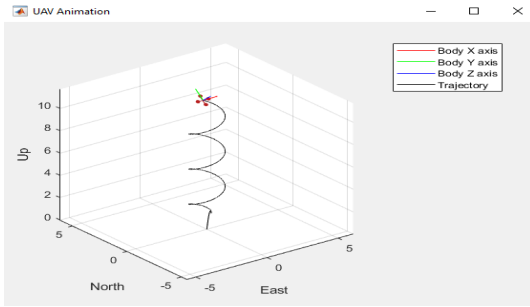


FIGURE 25. 3D animation for helical trajectory using UAV toolbox with GAFPID controller.

TABLE 6. Comparison of controllers using ITAE criteria for rectangular trajectory(undisturbed condition).

States	Fuzzy (ITAE)	FPID (ITAE)	GAFPID (ITAE)
x	21.42	18.44	18.00
y	27.67	46.82	46.82
z	12.97	1.86	1.33
ϕ	41.43	12.73	12.00
θ	159.20	10.02	9.27
ψ	1.95	1.88	1.73

TABLE 7. Comparison of controllers using ITAE criteria for rectangular trajectory tracking in the presence of input disturbances and parameter variation.

States	Fuzzy (ITAE)	FPID (ITAE)	GAFPID(ITAE)
x	21.55	18.11	18.01
y	29.62	89.48	89.46
z	16.56	2.902	2.487
ϕ	41.52	12.81	11.99
θ	160.4	10.08	9.3
ψ	1.96	1.89	1.74

VII. COMPARISON OF CONTROLLERS FOR RECTANGULAR TRAJECTORY BASED ON PERFORMANCE CRITERIA

Control system design plays a pivotal role in achieving precise trajectory tracking, especially when dealing with complex and continuous trajectories, such as helical and rectangular trajectories. The efficiency of controllers in navigating disturbances and variations in parameters while maintaining optimal performance is paramount. This study undertakes a comprehensive comparison of three controllers-Fuzzy controller, Fuzzy Proportional-Integral-Derivative (FPID), and a proposed controller, GAFPID (GA optimized Fuzzy PID controller) based on performance criteria in tabular form described in Tables 6 and 7 in undisturbed conditions and disturbed (with input disturbance and variation in parameters) respectively.

Upon analyzing the Tables comparing controllers under undisturbed 6 and disturbed 7 conditions for rectangular trajectory, notable trends emerge.

A. PERFORMANCE ANALYSIS

The performance of the proposed GAFPID controller is analyzed and compared with the Fuzzy and FPID controllers based on the ITAE (Integral of Time-weighted

Absolute Error) criteria under both undisturbed and disturbed conditions. The analysis is presented as follows:

B. UNDISTURBED CONDITIONS

1) POSITION STATES (X, Y, Z)

• GAFPID Controller:

- Demonstrates the best performance for x and z states, achieving lower ITAE values (18.00 and 1.33, respectively) compared to Fuzzy and FPID controllers.
- For the y state, GAFPID performs on par with FPID (46.82) but is inferior to Fuzzy (27.67), indicating limitations in lateral control.

• FPID Controller:

- Performs slightly worse than GAFPID for the x state but significantly outperforms Fuzzy for the z state.

• Fuzzy Controller:

- Shows competitive results for the x and y states but is significantly inferior for z , with a much higher ITAE value of 12.97.

2) ORIENTATION STATES (ϕ, θ, ψ)

• GAFPID Controller:

- Outperforms both Fuzzy and FPID controllers for all orientation states, particularly ϕ and θ , with ITAE values of 12.00 and 9.27, respectively.
- Achieves the lowest ITAE for ψ (1.73), demonstrating superior orientation stability.

• FPID Controller:

- Performs well in orientation control but is consistently outperformed by GAFPID.

• Fuzzy Controller:

- Exhibits the highest ITAE values for all orientation states, particularly for θ (159.20), indicating poor orientation control.

C. DISTURBED CONDITIONS (INPUT DISTURBANCES AND PARAMETER VARIATIONS)

1) POSITION STATES (X, Y, Z)

• GAFPID Controller:

- Maintains robust performance under disturbances, with minimal increases in ITAE values for x (18.01) and z (2.487), demonstrating effective disturbance rejection.
- Similar to undisturbed conditions, its y state performance (89.46) matches FPID but is inferior to Fuzzy (29.62), highlighting limitations in lateral disturbance handling.

• FPID Controller:

- Performs slightly worse than GAFPID in z state but matches GAFPID in y . Improves its x performance under disturbances compared to undisturbed conditions.

• Fuzzy Controller:

- Excels in y state under disturbances but struggles in z and x states, where it exhibits significantly higher ITAE values.

2) ORIENTATION STATES (ϕ, θ, ψ)

- **GAFPID Controller:**
 - Continues to demonstrate excellent disturbance rejection for orientation states. ITAE values remain the lowest for ϕ (11.99), θ (9.3), and ψ (1.74).
- **FPID Controller:**
 - Performs adequately but is outperformed by GAFPID across all orientation states.
- **Fuzzy Controller:**
 - Performs poorly for orientation states under disturbances, maintaining the highest ITAE values across the board.

D. SUMMARY OF LIMITATIONS

1) LIMITATIONS OF GAFPID

- **Lateral position Control (y) state:** The GAFPID controller consistently underperforms in the y state compared to Fuzzy, under both undisturbed and disturbed conditions.
- **Complexity:** The optimization-based design of GAFPID requires significant computational resources, which may limit real-time applicability.

VIII. CONCLUSION AND FUTURE WORK

The performance analysis of the GAFPID controller in comparison to Fuzzy and FPID controllers demonstrates that GAFPID is a robust and effective control strategy for trajectory tracking tasks. In undisturbed conditions, GAFPID achieves superior performance in most output states, particularly for altitude control (z) and orientations (ϕ, θ, ψ) control, as indicated by its lower ITAE values. Under disturbed conditions, GAFPID maintains its robustness and outperforms both Fuzzy and FPID controllers in rejecting disturbances and adapting to parameter variations. However, its consistent underperformance in lateral position control (y) highlights a key limitation. Fuzzy control exhibits strong lateral position control capabilities, particularly under disturbed conditions, but its poor performance in other output states and sensitivity to parameter variations make it less reliable overall. FPID achieves a balance between simplicity and performance, but it is generally outperformed by GAFPID in critical states like altitude (z) and pitch (θ) dynamic control.

A. RECOMMENDATIONS FOR FUTURE WORK

Based on the findings of this study, the following areas are recommended for future exploration:

- **Enhancing Lateral Position (y) Control:** Investigate optimization techniques or hybrid approaches, such as combining GAFPID with Fuzzy control, to improve performance in y -state control.
- **Dynamic and Adaptive Controllers:** Develop adaptive gain scheduling methods to enhance disturbance

rejection and adaptability in real-time applications. For instance, model reference adaptive control (MRAC) can address the computational complexity of GAFPID using simplified optimization techniques or real-time tuning algorithms.

- **Validation in Complex Environments:** Extend the evaluation of GAFPID to more complex trajectories, multi-agent systems, or scenarios with time-varying disturbances to assess scalability and robustness.
- **Hybrid Control Architectures:** Explore the integration of GAFPID with model-based or learning-based controllers (e.g., MRAC) to leverage their strengths for improved performance and flexibility.

Addressing these challenges will further enhance the applicability and reliability of GAFPID controllers in practical applications for quadrotor UAVs.

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interests include UAVs, intelligent control, adaptive control, optimal control, and cyber-physical systems.

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